AI5707 – Pattern Recognition and Machine Learning
Assignment 8: LDA, KNN, SVM, DTW

Due: Nov 10 2026

1 Objective

The objective of this assignment is to work with different data sets and classifiers. You are expected to draw inferences w.r.t dataset and the classifier used.

2 Datasets

You will work the following datesets:

1. Synthetic dataset
2. MNIST dataset
3. TI digits dataset (sequential)
4. Image dataset

3 Problems

Each teams works on the dataset relevant to the particular team. You build the following classifiers:

a GMMs

b LDA

c K-NN

d SVM

e DTW – digit dataset

1. Compare and contrast the accuracies for each of these dataset wrt to the classifier used. Comment on your results.
2. Perform PCA on the input feature space, and repeat the above experiment with features as projections across top K PCA dimensions.